1) Fly-Sky :

Binding the Transmitter and Receiver

To bind the Flysky FS-i6 transmitter with the FS-iA6B receiver:

1. Insert the **bind jumper** between the **BIND (signal)** pin and **GND** pin on the receiver *(this is usually the last pin marked "BIND" or B/VCC)*
2. Power the receiver by supplying **5 V and GND** from:
    - Flight controller, **or**
    - External BEC / power module
3. The receiver LED will start **flashing**, indicating bind mode
4. Now **press and hold the BIND KEY** on the FS-i6 transmitter
5. While holding the bind key, **power ON the transmitter**
6. Once the receiver LED turns **solid**, binding is successful
7. Power off everything and **remove the bind jumper**

Communication Modes: PWM vs PPM

The FS-iA6B receiver supports **two signal output modes**:

▸ **PWM (Pulse Width Modulation)**

- Each channel has a **separate signal wire**
- For 6 channels → **6 signal wires**
- More wiring, less convenient

▸ **PPM (Pulse Position Modulation)**

- **All channels combined into a single signal wire**
- Uses **PPM / CH1 pin** on the receiver
- Cleaner wiring and easier flight-controller setup

Enabling PPM Output on FS-i6

To enable PPM output:

1. **Long press OK** → enter **System**
2. Go to **RX Setup**
3. Enable **PPM Output** → **ON**
4. Enable **AFHDS** → **ON**
5. **Long press CANCEL** to save and exit
   *by default PWM is activated.

Transmitter Setup & Channel Configuration

1. **Long press OK** → enter **Setup**
2. **Reverse Channels**
    - Used if stick direction is inverted
    - Example: forward stick → quad moves backwards
3. **Auxiliary Channels**
    - FS-i6 supports **only 2 AUX channels** (CH5 & CH6)
    - Assign switches or knobs to:
        - **Channel 5**
        - **Channel 6**
    - Used for:
        - Flight modes
        - Arm / Disarm
        - Payload release, etc.
4. **Long press CANCEL** to save

2) Radio - Master:

Unlike Flysky, modern radio systems use **digital packet-based communication** instead of simple analog PWM/PPM signals.
In our lab, we use **RadioMaster** transmitters running **ExpressLRS (ELRS)**.

Transmitters available in the lab

- RadioMaster Pocket
- RadioMaster TX12
- RadioMaster TX16S

Receivers available in the lab

- RP1
- RP2
- RP4

All these transmitters and receivers **work in the same way** under ExpressLRS.

Checking Receiver Firmware (Wi-Fi Mode)

Before flashing or binding, we must identify the **receiver model and firmware**.

Steps:

1. **Power ON the receiver only**
2. Keep the transmitter **OFF**
3. Wait for about **60 seconds**
4. The receiver automatically enters **Wi-Fi mode**

On your laptop, connect to Wi-Fi network:
`ELRS RX`
`Password: expresslrs`

5. A webpage opens automatically (or go to `10.0.0.1`)
6. Under the **ExpressLRS logo**, note:
    - Receiver **model name**
    - Current **firmware version**

Flashing ExpressLRS Firmware on Receiver

Step A: Install ExpressLRS Configurator

- Download **ExpressLRS Configurator**
- Open it on your laptop

Step B: Select Target & Firmware

1. Go to **Configurator** tab
2. In **Target**, select the **exact receiver model** you saw on the webpage
3. Choose a **compatible firmware version**
    - ⚠ Not all latest versions work on all receivers
    - Limited flash memory → some versions will fail
    - This is **trial-and-check**, but use stable releases

Step C: Flash Settings

- Flashing method: **Flash over Wi-Fi**
- Regulatory Domain: **Standard**

- Set a **Binding Phrase**
    - Must be **unique**
    - Must be **exactly the same** for transmitter and receiver

Step D: Build & Upload

1. Click **Build**
2. Firmware file will be generated
3. Reconnect to **ELRS RX Wi-Fi**
4. Open the webpage → **Update**
5. Upload the downloaded firmware file
6. Receiver will flash and reboot automatically

Flashing ExpressLRS on RadioMaster Transmitter

Steps:

1. Power ON transmitter
2. Press **SYS / System**
3. Go to **ExpressLRS (LRS)**
4. Scroll to **Wi-Fi Connectivity**
5. Enable **Wi-Fi**
6. On laptop, connect to transmitter Wi-Fi
7. Open webpage → note
    - Transmitter model
    - Current firmware

Build Firmware (Same as Receiver)

- Select **same firmware version**
- Select **same binding phrase**
- Target must match **transmitter model**
- Click **Build**
- Upload firmware via **Update** page after connecting to wifi

Binding (Automatic with Binding Phrase)

Once both transmitter and receiver have:

- Same **firmware version**
- Same **binding phrase**

Final Step:

1. Power OFF both
2. Power ON transmitter
3. Power ON receiver

➡ **They bind automatically**
➡ No bind button or jumper required

RadioMaster Channel & Switch Configuration

After flashing and binding, we configure channels on the **RadioMaster** transmitter.

Mixes Page (Channel Mapping)

1. Press the **MODEL** button
2. Navigate to **Mixes**
3. Channels **CH1–CH4** are already fixed:
    - Roll
    - Pitch
    - Throttle
    - Yaw
4. From **CH5 onward**, auxiliary channels can be configured

Assigning Auxiliary Channels (CH5+)

1. Select **CH5** (or any higher channel)
2. **Long press OK** → Edit
3. Set:
    - **Source**
        - Click on Source
        - Toggle the desired switch (SA / SB / SC / SD)
        - That switch gets assigned automatically

- **Weight**
    - `+100` → normal direction
    - `-100` → reverses the channel

This allows:

- Switch-based flight modes
- Arming / disarming
- Payload drop
- Any other auxiliary control

RadioMaster supports **up to 32 channels**,
but **Pixhawk** typically requires only **4–6 auxiliary channels**.

Special Functions (Audio Feedback)

Special Functions are used for **pilot feedback**, not control.

Steps:

1. Go to the **Special Functions page** after clicking the **MDL button.**
2. Select a switch (example: **SA**)
3. Choose an action:
    - Play Track
    - Play Sound
4. Assign a sound like:
    - “Armed”
    - “Disarmed”
    - “Mode changed”

Example:

- If **SA** is your arming switch
- You can assign:
    - **SA↑ → “Armed”**
    - **SA↓ → “Disarmed”**

This gives instant **audio confirmation** during flight.